Causal Forecast-Conditioned BS-RoFormer for Ocean Sound Separation

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***Abstract*—We evaluate FC-BSR with one fixed main setting: a causal GRU, FiLM, a BS-RoFormer separator, and 5-s windows. Each supervised example is (H_t, N_t, s_t, x_t): twelve immediately preceding 5-s background features whose annotated whale-call intervals are masked, an annotation-clean real 5-s ocean segment, a synthetic marine event, and their exact mixture. The available recordings provide 146.4 h of call-free background, with 37.3 h eligible for training. The 12-step GRU reduces held-out forecast MAE from 2.111 to 1.926 dB (8.8%) against persistence. A six-way, matched-condition audit on 4,096 held-out mixtures finds 9.631 dB target SI-SDRi for both the true BS-RoFormer and Zero-FiLM, 9.497 dB for GRU-Forecast FiLM, and 9.535 dB for Oracle-FiLM using the actual current 30-band $E(N_t)$. Oracle minus BS-RoFormer is -0.096 dB (day-block 95% CI $[-0.238, 0.034]$): this 30-band condition has no demonstrated average benefit. A separate full-waveform oracle copy check learns $x_t - N_t$ on the same held-out mixtures, showing that the compressed condition is not a full-information oracle. With the copy interface fixed, an information ladder retains a positive day-block gain at 128 equal-width frequency groups but not at 64; 30 geometric bands lose a confirmed gain even at the finest tested time grid. A trained 128-group time-resolved Oracle residual separator reaches 16.767 dB on the same held-out mixtures versus 9.631 dB for BSR (paired $+7.136$ dB, day-block CI $[6.970, 7.282]$); no forecast is used in this Oracle test. A separate 13-family study compares other conditioning modes as ablations. On untouched real calls, where no isolated stem exists, a frozen event classifier finds no clear semantic advantage over a matched zero-condition FiLM control. We therefore support fully supervised separation on annotated-clean backgrounds, but the present 30-band forecast/FiLM interface has no established separation gain.**

***Index Terms*—ocean acoustics, source separation, soundscape forecasting, BS-RoFormer, weak-reference evaluation**

I. Introduction

Ocean backgrounds vary with shipping, weather, biology, and season [1]. A mixture-only separator must infer both an unexpected event and this changing background from the same observation. We instead ask whether the preceding soundscape can supply a causal prior. Our main data tuple is

$$\begin{aligned} &(H_t, N_t, s_t, x_t), \quad x_t = N_t + s_t, \\ &H_t \xrightarrow{\text{GRU}} \widehat{S}_t, \qquad x_t \xrightarrow[\widehat{S}_t]{\text{FiLM+BSR}} (\hat{s}_t, \hat{N}_t), \end{aligned} \tag{1}$$

where H_t excludes annotated call intervals and N_t is a real current no-call segment. Training has exact stems, while the separator never receives N_t or s_t as input.

Time-domain separators [2], [3], attention models [4], [5], and band-split systems [6], [7] provide strong mixture-only backbones. Hybrid Transformers and MixIT broaden the architecture and supervision regimes [8], [9], but do not test a forecast of the upcoming background. FiLM, cross-attention, and cross-modal separation [10]–[12] motivate three interfaces. Forecasting baselines range from recurrent to linear and patch-based models [13]–[16].

Our contributions are: (1) FC-BSR,[1] with a single fixed main setting (GRU+FiLM+BS-RoFormer, 5 s); (2) a reproducible annotation-aware construction of 146.4 h of current call-free ocean background and event-masked 5-s histories; (3) a 13-family broad event generator with seven whale/vessel families in the main DCLDE experiment; and (4) a frozen-classifier evaluation on untouched real calls against a zero-condition FiLM control. TOL-domain, conditioning-mode, time-scale, and unannotated-background strength studies are reported as ablations rather than as the primary evidence.

II. Method

A. Forecast and conditional separator

The main history $H_t \in \mathbb{R}^{12\times 30}$ contains 30-band log-power features computed on twelve preceding 5-s windows; annotated calls and their 2-s guards are excluded. Features are normalized using training data only. A residual GRU predicts

$$z_t = \mathrm{LN}(\mathrm{GRU}(H_t)), \quad \widehat{S}_t = S_{t-1} + g_\psi(z_t). \tag{2}$$

Residual initialization represents persistence exactly; a learned predictor is retained only if validation MAE improves upon persistence [17].

The main FC-BSR maps the mixture STFT to band tokens X_k and applies FiLM before the axial Transformer:

$$\widetilde{X}_k = (1 + \tanh \gamma_k) \odot X_k + \beta_k. \tag{3}$$

The separator predicts an event stem; the ocean stem is the mixture residual. Training combines waveform L_1 and multi-resolution complex-STFT losses. Concatenation and cross-attention are alternative conditioning ablations, not the deployed FC-BSR configuration.

The main-domain information audit holds the separator backbone, mixtures, initialization, and training rule fixed. A true BS-RoFormer bypasses the condition injector; Zero-FiLM retains the FiLM branch and the same frozen GRU, but replaces its output by $c = 0$. Shuffled-FiLM uses $E(N_j)$ from a deterministic half-cycle offset in the clean-window index of the same split ($j \neq t$); on the held-out test, this selects a different day. Persistence-FiLM uses S_{t-1}, and Forecast-FiLM uses the

[1]Code and reproduction instructions: https://github.com/D-eval/ocean_noise_pred_splite.

frozen GRU output. Oracle-FiLM receives $S_t^{\text{oracle}} = E(N_t)$ computed from the actual current clean background after any shared mixture scaling. We call this the *30-band Oracle-FiLM*: E removes temporal structure and phase, so this row bounds neither a full-background condition nor the value of perfect forecasting with a richer representation. It is available only in the controlled experiment.

B. Marine-aware source distribution

The broad generator contains 13 families: harmonic, linear chirp, inharmonic, impulse train, frequency-modulated tone, band noise, whale upsweep, downsweep, arch, S-curve, pulse packet, vessel tonal, and vessel cavitation. Whale proxies use one to three localized events with variable duration, curvature, partials, taper, and flutter. Vessel-tonal and narrow-band cavitation components draw a 5–20 Hz low-frequency-oscillator amplitude envelope with probability 0.8; their broadband component is not modulated. Training SNR is uniform on $[-18, 12]$ dB. This distribution is intended to cover contour and modulation patterns, not to claim species identity.

We quantify spectral displacement with directional $D_{\mathrm{KL}}(p\|q)$ and finite symmetric Jensen–Shannon (JS) divergence [19]; spectral entropy and source–ocean overlap are reported separately. Larger KL/JS expands coverage but is not treated as semantic diversity. The DCLDE main experiment samples only its five whale and two vessel families; all six main-domain models see the same target distribution.

C. Main data construction and secondary ablations

The main dataset uses day-disjoint 2-kHz DCLDE 2013 recordings and whale-call annotations [20], [21]. Across 168 h of WAV, excluding annotated calls and a 2-s guard on each side yields 146.4 h of annotation-clean 5-s windows. For each target, the 12 immediately preceding 5-s blocks from the same WAV form a 30-band history; a target is eligible only when all history blocks are outside annotated calls and their 2-s guards. This avoids temporal averaging over an hour and forbids histories from crossing WAV boundaries. This leaves 94.5 h eligible overall, split into 37.3/9.7/47.6 h for train/validation/test. For each current clean N_t, a seven-family whale/vessel synthetic event s_t is mixed to form $x_t = N_t + s_t$. The main split is by recording day, and the GRU receives 12 event-masked features from the immediately preceding 5-s windows.

Secondary ablations use NOAA–Navy SanctSound hourly TOL products [18]. MB03 deployments 01–02/03/04 are train/validation/test. Random-phase inversion creates controlled spectral surrogates, not recovered audio. Concatenation, cross-attention, KL/JS coverage, and the one-hour history-resolution comparison are confined to this domain. The separate 42-h SanctSound WAV strength sweep also lacks event annotations, so it is a useful controlled real-background ablation but is not counted as clean-background primary evidence.

Complete-reference tiers report SI-SDR and SI-SDRi using scale-invariant projection [22], [23]. All uncertainty intervals are seeded paired bootstraps, blocked by recording day for the main test. The event classifier is frozen before any real-event evaluation. Current test features are used only as targets for the synthetic full-reference task, never as GRU input; all histories end before t.

TABLE I
CONDITIONING-MODE ABLATION ON TOL-DERIVED SPECTRAL SURROGATES (DB).

Injection	Val. SI-SDRi	Test SI-SDRi	Paired Δ_0
None	4.864	4.059	0
Concatenation	5.343	4.432	+0.623
FiLM	**5.419**	**4.468**	**+0.692**
Cross-attention	5.128	4.261	+0.331

III. Experiments and Results

A. Conditioning-mode ablation on spectral surrogates

The primary architecture is fixed to FiLM. Table I compares it with the alternative interfaces on controlled 5-s spectral-surrogate examples; on 832 fixed-0-dB pairs (64 per family), FiLM improves 4.850 to 5.541 dB SI-SDRi. Band-wise scale and bias directly modulate existing band evidence. Concatenation must compress doubled features, while cross-attention must learn alignment from coarse predicted spectra under the same budget.

FiLM gains 0.869, 0.835, 0.919, 0.906, and 0.806 dB on the five whale proxy families. Vessel tonal is the hardest family: the conditioned absolute SI-SDRi is -8.295 dB, despite a 0.320-dB relative gain. Overall gain has weak correlation with JS ($r = -.057$) and entropy ($r = .122$), so increasing divergence broadens support but does not predict usefulness.

B. Primary fully supervised clean-background separation

The main FC-BSR receives twelve event-masked preceding 5-s features and a current clean 5-s DCLDE segment; the seven-family event is added only after the clean background is selected. The model therefore sees x_t and H_t, while the exact N_t and s_t are supervision. The training split contains 37.3 h of eligible clean current audio. The 12-step GRU forecasts the next 5-s band spectrum; validation MAE is 1.926 dB versus 2.111 dB for persistence (8.8% lower). The older two-row comparison labeled a Zero-FiLM checkpoint as "BS-RoFormer." We retain its exact trained waveform mapping as the Zero control, attach the same frozen GRU in a bypassed branch, and train a true injector-bypassed BS-RoFormer. All six rows in Table II use the same mixtures, transferred backbone initialization, optimizer budget, validation selection, and 4,096 test examples from three held-out days. Figure 3 stratifies these same samples by input SNR.

The architecture effect, Zero minus BS-RoFormer, is -0.0002 dB (95% CI $[-0.0068, 0.0088]$). Forecast minus BS-RoFormer is -0.134 dB ($[-0.327, 0.125]$); Oracle minus BS-RoFormer is -0.096 dB ($[-0.238, 0.034]$). Oracle minus Forecast is only $+0.039$ dB ($[-0.090, 0.155]$). Shuffling and persistence also fail to show a positive average over Zero. Thus the matched controls do not establish that temporal forecast information improves this separator. Oracle's higher validation

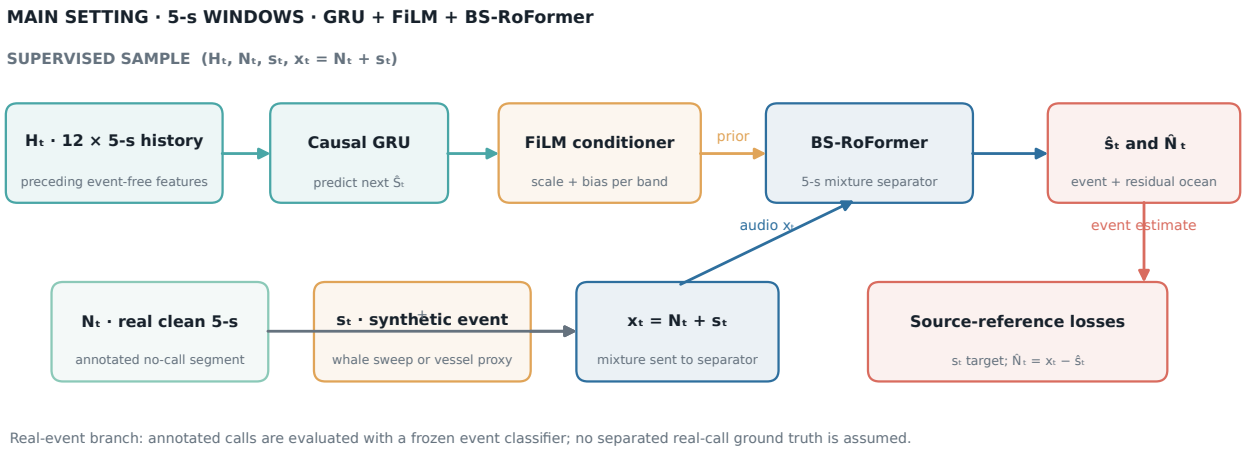


Fig. 1. The fixed main setting and its supervised data tuple. Event-masked history drives the GRU forecast; real annotated-clean background and a synthetic event form the current mixture. FiLM conditions the BS-RoFormer and the event estimate is supervised and the ocean estimate is its mixture-consistent residual.

score (11.372 versus 10.946 dB for BS-RoFormer) does not transfer to a clear held-out gain. With only three test days, the intervals leave small effects unresolved; the 30-band oracle supplies more information, but separately optimized FiLM models do not form a guaranteed performance ordering.

a) Full-waveform oracle copy check.: To test whether losing information in E explains the negative 30-band result, we freeze the same trained BS-RoFormer and fit a pointwise, three-coefficient head on 8,192 training mixtures:

$$\hat{s}_t = w_b \hat{s}_{\mathrm{BSR}}(x_t) + w_x x_t + w_c c_t. \tag{4}$$

The head starts at $(1, 0, 0)$, exactly reproducing BSR, and its waveform-MSE global optimum is solved by least squares. We fit the *same* head with zero, wrong-time, and actual current background waveforms; validation and testing use 4,096 examples each, with the test IDs, SNRs, mixture scores, and frozen BSR outputs checked against Table II. For $c_t = N_t$, the learned weights are $(0, 1, -1)$ to numerical precision: the head copies the exact answer $x_t - N_t = s_t$. The maximum test waveform error is 5.96×10^{-8}; the finite SI-SDRi of 80.755 dB is limited by the metric's numerical floor. Zero and wrong-time heads receive no useful background identity and do not improve SI-SDRi (Table III). This is an intentionally noncausal information/path sanity check with a different interface, not an architecture-matched FiLM ablation or evidence that the 30-band forecast can recover a phase-aligned background waveform.

b) Oracle information ladder.: We next reduce the information in the *current* N_t while keeping the frozen BSR, the three-coefficient head, training examples, and 4,096 matched test mixtures fixed. A 512-point Hann STFT at 2 kHz has 257 frequency bins on a 64-ms hop. We average background power across either adjacent equal-width groups (128, 64, 32) or geometric 20–950-Hz groups (128, 64, 48, 30), and separately pool frames over 0.25, 1, or 5 s. The retained magnitude is inverted with either the actual N_t phase or the mixture x_t phase. Geometric groups absorb edge bins; only 95 of the nominal 128 geometric groups contain an STFT bin. The 30-band/5-s rung is analogous to, but not numerically identical with, the original $E(N_t)$.

Figure 2 identifies where this *copy path* ceases to show a held-out gain. Exact amplitude with mixture phase still adds 11.164 dB over BSR (day-block CI $[10.889, 11.594]$). With equal-width groups, 128 retains $+2.067$ dB ($[0.969, 3.386]$), while 64 has -0.623 dB ($[-1.404, 0.259]$): the confirmed gain disappears between these levels. For geometric bands, nominal 128 already gives -0.529 dB ($[-1.257, 0.319]$), and 30 per-frame bands give -0.672 dB ($[-1.323, 0.086]$). Retaining the true phase at 30 per-frame bands gives only $+0.117$ dB ($[-0.671, 1.084]$). By contrast, keeping all 257 bins but pooling power to 1 s still yields $+0.585$ dB with mixture phase ($[0.285, 1.132]$); the 5-s version yields $+0.249$ dB ($[-0.044, 0.495]$). Thus this interface is especially sensitive to frequency averaging and the grouping rule, while time averaging alone is less destructive. These are sufficiency results for the fixed waveform copy head, not information-theoretic limits or new FiLM results.

c) Trained high-resolution Oracle residual separator.: We next test the representation in an actual BS-RoFormer, rather than only a frozen linear copy head. The actual current N_t and mixture x_t are transformed with a 512-point, 128-hop STFT. Mean power in 128 equal-width frequency groups yields time-resolved $\log|B_t|$; the explicit innovation is $R_t = \log|X_t| - \log|B_t|$ on the same grid. These maps are interpolated to the BSR frame grid and projected into its mixture-derived band tokens before axial attention. This zero-initialized residualized $[X_t, B_t, R_t]$ input uses neither FiLM nor a forecaster; its initial waveform output exactly matches ordinary BSR (maximum difference zero).

Both models use the same day-disjoint DCLDE mixtures,

TABLE II

MAIN DCLDE TEST ON 4,096 MATCHED ANNOTATION-CLEAN 5-S BACKGROUNDS PLUS SYNTHETIC EVENTS. VALUES ARE TARGET SI-SDRI (DB); BRACKETS ARE PAIRED 95% BOOTSTRAP INTERVALS OVER HELD-OUT RECORDING DAYS. THE 30-BAND ORACLE ALONE SEES $E(N_t)$.

Model	Validation	Test	Δ vs. BSR [95% CI]	Δ vs. Zero [95% CI]
BS-RoFormer	10.946	9.631	0	$+0.0002$ $[-0.009, 0.007]$
Zero-condition FiLM	10.967	9.631	-0.0002 $[-0.007, 0.009]$	0
Shuffled-condition FiLM	10.609	9.526	-0.105 $[-0.247, 0.020]$	-0.105 $[-0.256, 0.026]$
Persistence-condition FiLM	11.205	9.524	-0.107 $[-0.239, 0.083]$	-0.107 $[-0.232, 0.084]$
GRU-Forecast FiLM (FC-BSR)	11.213	9.497	-0.134 $[-0.327, 0.125]$	-0.134 $[-0.320, 0.125]$
30-band Oracle-FiLM, $E(N_t)$	11.372	9.535	-0.096 $[-0.238, 0.034]$	-0.095 $[-0.231, 0.035]$

TABLE III

FULL-WAVEFORM ORACLE COPY CHECK ON THE SAME 4,096 DCLDE TEST MIXTURES. TARGET SI-SDRI AND PAIRED DAY-BLOCK Δ AGAINST THE FROZEN BSR (DB).

Frozen BSR + fitted head	SI-SDRi	Δ [95% CI]
BSR; initial $(1, 0, 0)$	9.631	0
Zero waveform	9.231	-0.400 $[-0.483, -0.269]$
Wrong-time waveform	9.231	-0.400 $[-0.483, -0.269]$
Actual N_t waveform	80.755	$+71.124$ $[69.945, 72.081]$
Analytic $x_t - N_t$	80.755	$+71.124$ $[69.945, 72.081]$

TABLE IV

ABLATION: PREDICTOR-RESOLUTION DIAGNOSTIC ON SANCTSOUND BACKGROUND.

Step	Predictor	Cond.	Base	Gain [95% CI]
5 s	GRU	29.753	28.527	1.226 [1.082,1.381]
30 s	Persist.	29.630	28.534	1.095 [0.935,1.265]
1 min	Persist.	29.661	28.533	1.128 [0.971,1.298]
10 min	Persist.	29.711	28.535	1.176 [1.036,1.326]

Cond./Base/Gain are target SI-SDRi in dB.

transferred backbone initialization, seed, batch order, 6,000 updates, AdamW/cosine schedule, loss, and validation-best checkpoint rule. The existing BSR selected step 5,300; the Oracle model selected step 6,000. On 4,096 matched test examples, BSR scores 9.631 and Oracle-Residual-BSR 16.767 dB target SI-SDRi. The paired improvement is $+7.136$ dB (recording-day 95% interval $[6.970, 7.282]$), with all three held-out days positive. The combined high-resolution prior and explicit innovation interface therefore benefits this controlled separator. Actual N_t is unavailable at deployment; this two-model comparison also does not isolate adapter capacity from the condition information.

C. Secondary ablations: time scale and target strength

The time-resolution diagnostic in Table IV is an ablation on continuous SanctSound audio, not the main annotation-clean data. Only its 5-s predictor beats persistence on validation (MAE 1.366 versus 1.386 dB); the other scales select persistence. The separate 42-h strength curve also uses unlabeled current background. It shows that forecast conditioning is strength-dependent in that domain and must not be read as a result on annotation-clean current segments.

The harder fixed-5-s experiment on unannotated SanctSound WAV compares Zero-condition FiLM, a current-spectrum oracle, causal forecast, and a time-shuffled condition on 1,152 paired test mixtures. This oracle belongs to the secondary data domain; Table II uses the actual annotation-clean DCLDE N_t. Averaged across training SNRs, the baseline scores 23.470 dB and Forecast-FC-BSR 22.871 dB ($\Delta = -0.599$, CI $[-0.851, -0.346]$); even the oracle is lower. This exposes conditioning/capacity effects that the simpler diagnostic misses. Figure 4 gives the more informative strength sweep. Forecast conditioning helps weak targets by 0.728, 0.643, 0.525, 0.366, and 0.157 dB at $-18, -15, -12, -9, -6$ dB, respectively, then loses from -3 dB onward. The shuffled condition also helps the weakest targets but collapses as SNR rises; therefore weak-target gain alone cannot prove that temporal forecast accuracy, rather than regularization or extra capacity, is responsible.

D. Real events without separated ground truth

Following learned reference-free metrics [24], [25] and semantic audio embeddings [26], [27], we train, calibrate, and freeze a binary event classifier (test AUROC .9723, AP .9755, F1 .9275). Mixture consistency imposes $\hat{N} = x - \hat{s}$. We restore $\hat{s}$ at gains $\{-60, -30, -24, -18, -12, -6, 0\}$ dB and test whether event probability recovers monotonically. Identity and all-to-event controls reject trivial solutions; changes outside annotated intervals audit preservation.

The same frozen 12-step event-masked forecaster used by the main setting supplies the causal history features. Because recent call activity is masked, 103 real test calls retain a full eligible 60-s history. Table V shows the untouched-call results: FC-BSR trends toward lower residual event probability and off-event change, while event-stem probability is nearly unchanged. The day-block intervals include zero for residual probability $\Delta = -.0731$ ($[-.2090, .0294]$), event-stem probability $\Delta = +.0022$ ($[-.0180, .0304]$), and monotonicity $\Delta = +.0178$ ($[-.0333, .0429]$). Off-event relative error decreases by .0248 ($[-.0456, -.0018]$). This small semantic set is suggestive but not decisive; separated-stem ground truth is still unavailable.

IV. DISCUSSION AND CONCLUSION

The main-domain 30-band Oracle is a diagnostic of the existing interface: even when FiLM receives $E(N_t)$ from the exact current clean waveform, its held-out mean does not exceed the mixture-only BS-RoFormer, and its paired interval spans zero. The full-waveform copy check succeeds exactly, so the current negative result cannot be interpreted as evidence that the true background lacks useful information. It

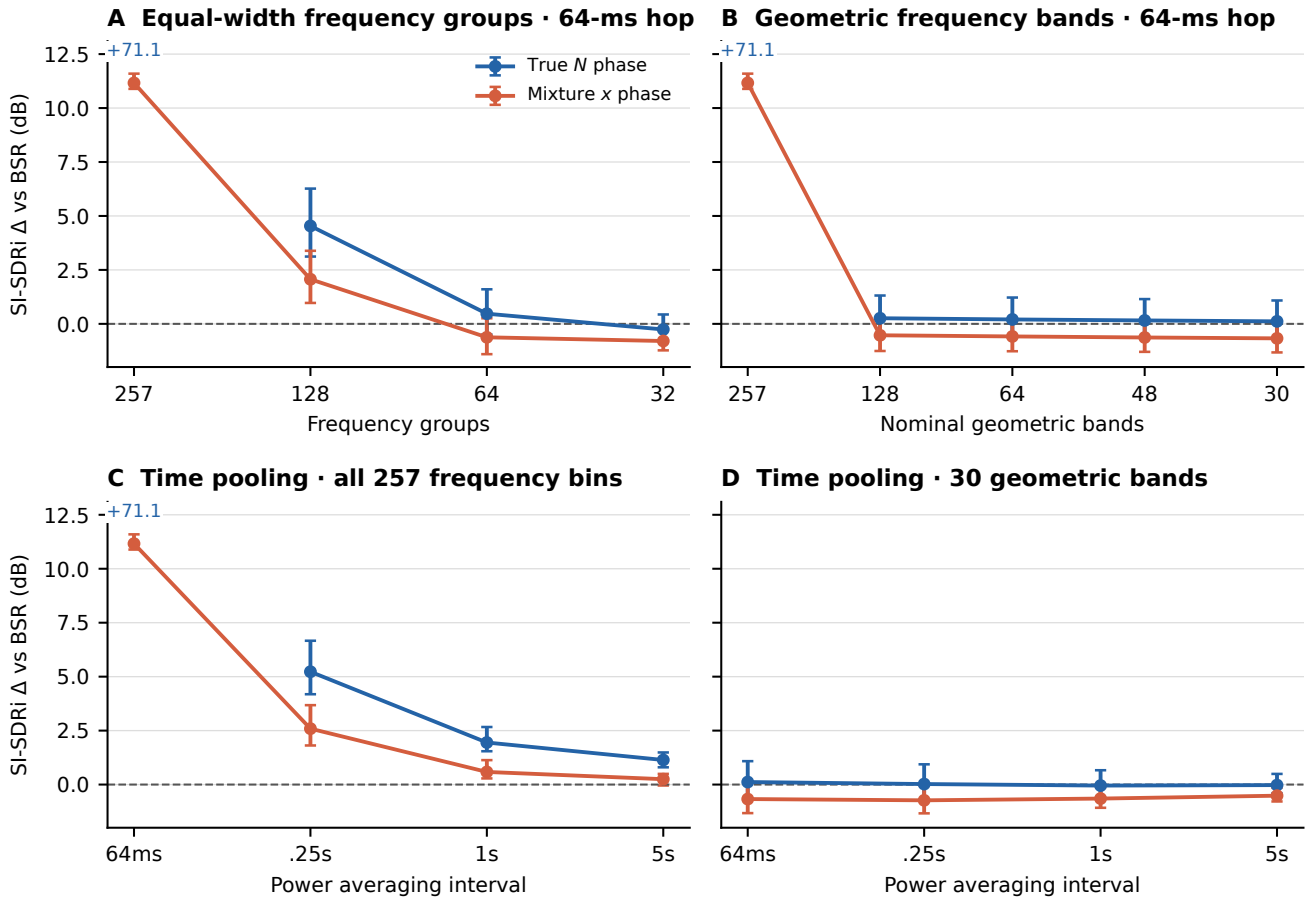


Fig. 2. Current-background Oracle information ladder on identical DCLDE test mixtures. Each point fits the same three-coefficient copy head on training mixtures. Vertical bars are paired 95% day-block intervals for SI-SDRi difference from frozen BSR. Full-resolution true-phase N_t gives +71.1 dB and is above the plotted range; mixture phase is a separate information loss. The threshold depends on frequency grouping and this copy interface.

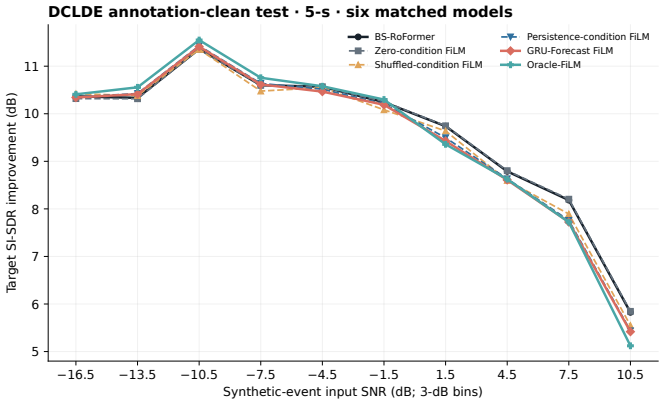


Fig. 3. Primary full-reference test on annotation-clean DCLDE backgrounds: mean target SI-SDRi in 3-dB input-SNR bins for all six matched methods. Table II gives day-block uncertainty for their overall means.

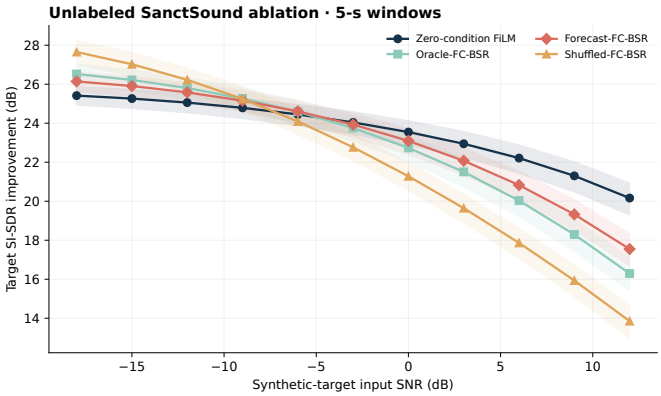


Fig. 4. Ablation on 5-s unlabeled SanctSound background: target-strength sweep. Lines show the four conditions; shaded bands are paired 95% intervals.

leaves the relative roles of 30-band compression and FiLM injection unresolved. The fixed-head ladder shows a clear representation bottleneck: even mixture-phase conditions retain a confirmed gain at 128 equal-width frequency groups or 1-s full-bin temporal pooling, whereas 64 equal-width groups and

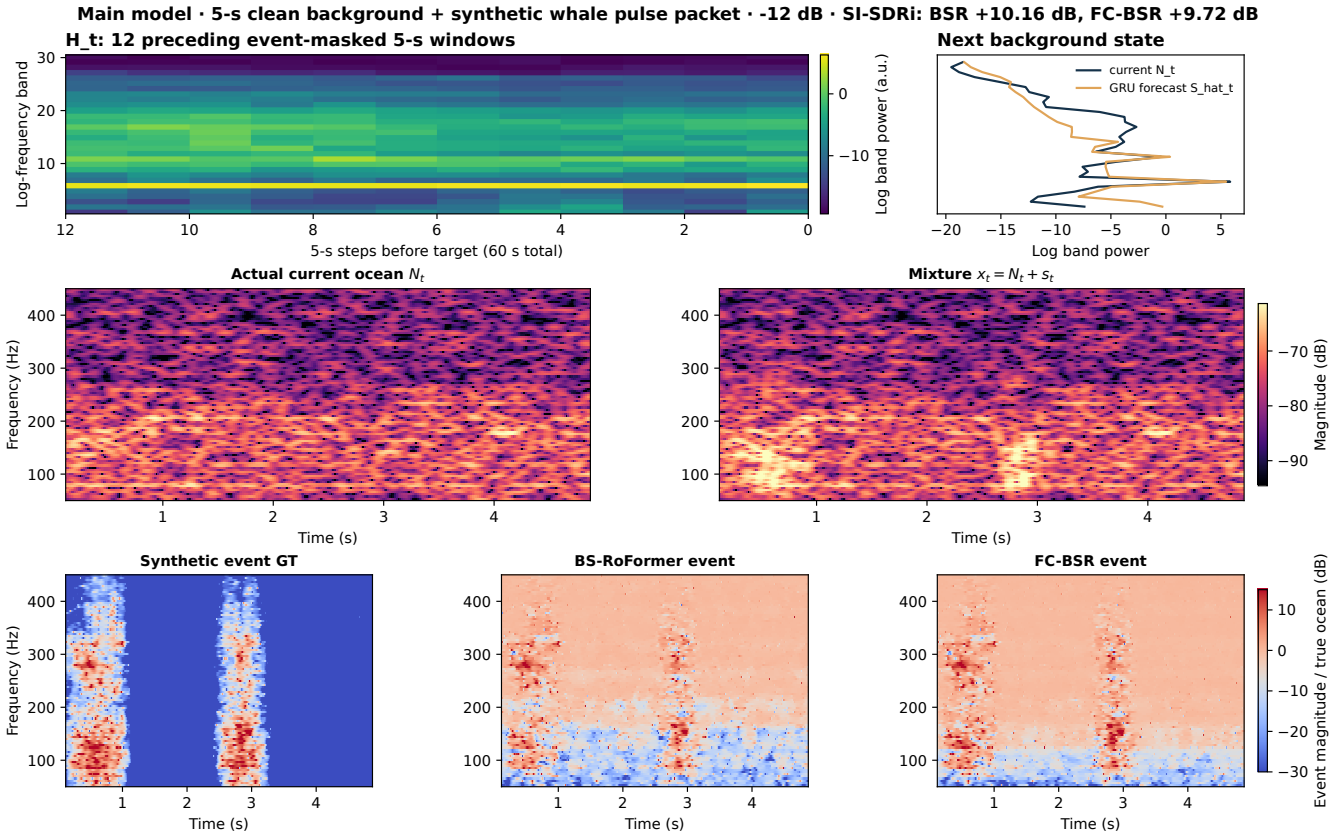


Fig. 5. Median-selected −12 dB synthetic whale pulse packet in an annotation-clean 5-s DCLDE recording. The ocean and mixture share an absolute magnitude scale; the lower row shows each event stem relative to the actual ocean magnitude on a common dB scale. Neither estimate fully removes background texture; aggregate results are in Fig. 3.

TABLE V
REFERENCE-FREE MATCHED COMPARISON ON 103 REAL CALLS WITH ELIGIBLE HISTORIES.

Method	$p(x)$	$p(\hat{N}) \downarrow$	$p(\hat{s}) \uparrow$	Mono.↑	Off-event↓
Zero-condition FiLM	.909	.786	.900	.937	.230
FC-BSR	.909	.713	.902	.955	.206

30 geometric bands do not. The original FiLM result cannot be assigned to this bottleneck alone because its interface differs. The trained 128-group Oracle-Residual-BSR gains +7.136 dB on matched held-out mixtures, showing that a high-resolution true background and explicit innovation can help the BSR backbone. The added adapter capacity and current-background Oracle access remain limitations of this feasibility test; a causal forecast has not been evaluated through this interface. Zero-FiLM is effectively tied with BS-RoFormer, while Forecast is no better than either Zero or Shuffled. The result does not isolate prediction error as the bottleneck; a five-second, 30-band static summary and its FiLM interface may be too coarse for the separator's time-frequency tokens. FiLM remains the best interface on controlled spectral surrogates, and the secondary unlabeled WAV sweep shows weak-target gains, but neither establishes a main-domain gain. No decisive forecast advantage appears on untouched whale calls. The main test has three held-out recording days, so small or site-dependent effects remain possible.

The local frontend is usable for inspection: on a CPU smoke test, one 5-s chunk ran at RTF 0.014 after history features were prepared; the interface displays forecasts and permits playback and download of both estimates. This does not establish field deployment. The next step is to test whether causal history can predict a sufficiently accurate time-frequency background for this same innovation interface. Further work should cover more sites and seasons and obtain isolated or stronger partial references for real events. Classifier-based

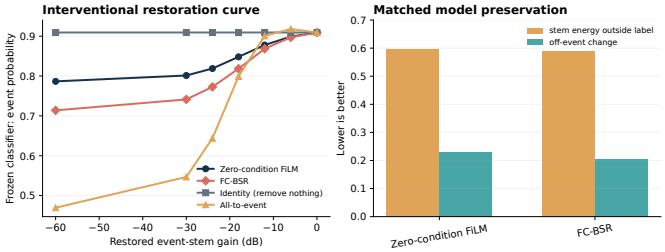


Fig. 6. Reference-free restoration curves and preservation audit for Zero-condition FiLM and FC-BSR. Both perform nontrivial semantic separation, but FC-BSR is not uniformly better on real calls.

restoration is evidence of semantic transfer, not a replacement for waveform fidelity.

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